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Reducing parasitic capacitance in field-effect transistors

Abstract

A semiconductor structure includes a semiconductor fin protruding from a substrate, an S/D feature disposed over the semiconductor fin, and a first dielectric fin and a second dielectric fin disposed over the substrate, where the semiconductor fin is disposed between the first dielectric fin and the second dielectric fin, where a first air gap is enclosed by a first sidewall of the epitaxial S/D feature and the first dielectric fin, and where a second air gap is enclosed by a second sidewall of the epitaxial S/D feature and the second dielectric fin.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/062,760, filed Dec. 7, 2022, which is a continuation of U.S. patent application Ser. No. 17/150,725, filed Jan. 15, 2021, now U.S. Pat. No. 11,532,502B2, which claims the benefits of U.S. Provisional Patent Application Ser. No. 63/002,489, filed Mar. 31, 2020, the entire disclosures of which are incorporated herein by reference.

BACKGROUND

(1) The semiconductor industry has experienced rapid growth. Technological advances in semiconductor materials and design have produced generations of semiconductor devices where each generation has smaller and more complex circuits than the previous generation. In the course of integrated circuit (IC) evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. But these advances have also increased the complexity of processing and manufacturing semiconductor devices.

(2) Three-dimensional field effect transistors, such as fin-like FETs (FinFETs) and gate-all-around (GAA) FETs (GAA FETs), have been incorporated into various memory and core devices to reduce IC chip footprint while maintaining reasonable processing margins. While methods of forming these FETs have generally been adequate, they have not been entirely satisfactory in all aspects. For example, lowering parasitic capacitance by introducing air gaps to device structures when the number of active regions (i.e., fins) reduces to two or less remains a challenge. Thus, for at least this reason, improvements in methods of fabricating FinFETs, GAA FETs, and the like are desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a flowchart of an example method for fabricating a semiconductor device according to various embodiments of the present disclosure.

(3) FIGS. 2 and 10A are planar top views of an example semiconductor device at various stages of the method shown in FIG. 1, according to various embodiments of the present disclosure.

(4) FIGS. 3, 4, 5A, 5B, 6, 7, 8, and 9 are cross-sectional views of the semiconductor device at various stages of the method shown in FIG. 1, according to various embodiments of the present disclosure.

(5) FIG. 10B is a three-dimensional perspective view of the semiconductor device shown in FIG. 10A, according to various embodiments of the present disclosure.

(6) FIGS. 11A, 12A, 13, 14, 15A, 15B, 16A, 16B, 16C, 16D, 17A, 17B, 18A, and 18B are cross-sectional views of the semiconductor device taken along line AA' of FIG. 10A and/or FIG. 10B at

various stages of the method shown in FIG. 1, according to various embodiments of the present disclosure.

(7) FIGS. 10C, 11B, 12B, 19A, 19B, 20A, and 20B are cross-sectional views of the semiconductor device taken along line BB' of FIG. 10A and/or FIG. 10B at various stages of the method shown in FIG. 1, according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

(8) The following disclosure provides many different embodiments, or examples, for implementing different features of the disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a feature on, connected to, and/or coupled to another feature in the present disclosure that follows may include embodiments in which the features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the features, such that the features may not be in direct contact. In addition, spatially relative terms, for example, “lower,” “upper,” “horizontal,” “vertical,” “above,” “over,” “below,” “beneath,” “up,” “down,” “top,” “bottom,” etc. as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) are used for ease of the present disclosure of one features relationship to another feature. The spatially relative terms are intended to cover different orientations of the device including the features.

(9) Furthermore, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range including the number described, such as within $\pm 10\%$ of the number described or other values as understood by person skilled in the art. For example, the term “about 5 nm” encompasses the dimension range from 4.5 nm to 5.5 nm. Still further, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(10) The present disclosure provides methods of reducing parasitic capacitance in field-effect transistors (FETs) including fin-like FETs (FinFETs) and multi-gate FETs such as gate-all-around (GAA) FETs. Each FinFET includes a three-dimensional channel region that engages with a gate structure, while each GAA FET includes a vertical stack of nanosheets, nanowires, or nanorods as channel regions interleaved with the gate structure. Some embodiments provide methods of lowering parasitic capacitance when a number of fins (or active regions) in a given FET structure is reduced to two or less. Embodiments of the present disclosure may be readily integrated into existing process flow for forming epitaxial S/D features in FinFETs, GAA FETs, and/or other suitable FETs.

(11) Referring to FIG. 1, a flowchart of method **100** of forming a semiconductor device **200** (hereafter simply referred to as the device **200**) are illustrated according to various aspects of the present disclosure. Method **100** is merely an example and is not intended to limit the present disclosure beyond what is explicitly recited in the claims. Additional operations can be provided before, during, and after method **100**, and some operations described can be replaced, eliminated, or moved around for additional embodiments of the method. Method **100** is described below in conjunction with FIGS. 2-20B, where FIGS. 2 and 10A illustrate planar top views, FIGS. 3-9, 10C-20B illustrate cross-sectional views, FIG. 10B illustrates a three-dimensional perspective view of the device **200**. Specifically, FIGS. 11A, 12A, and 13-18B illustrate cross-sectional views of the device **200** taken along line AA' as shown in FIGS. 10A and/or 10B, and FIGS. 11B, 12B, and 19A-20B illustrate cross-sectional views of the device **200** taken along line BB' as shown in FIGS. 10A and/or 10B. The device **200** may be an intermediate device fabricated during processing of an IC, or a portion thereof, that may comprise static random-access memory (SRAM) and/or other logic circuits, passive components such as resistors, capacitors, and inductors, and active components such as FinFETs, GAA FETs, metal-oxide semiconductor field effect transistors

(MOSFETs), complementary metal-oxide semiconductor (CMOS) transistors, bipolar transistors, high voltage transistors, high frequency transistors, and/or other transistors. The present disclosure is not limited to any particular number of devices or device regions, or to any particular device configurations. Additional features can be added to the device **200**, and some of the features described below can be replaced, modified, or eliminated in other embodiments of the device **200**.

(12) At operation **102**, referring to FIGS. **2** and **3**, method **100** forms a device region **203** over a semiconductor substrate (hereafter referred to as the “substrate”) **202**, where the device region **203** is configured to provide one or more FETs. The device region **203** includes at least one semiconductor fin protruding from the substrate **202** and oriented lengthwise along the X direction. In some embodiments, as depicted herein, the device region **203** includes an array of multiple semiconductor fins **204** and **204'** spaced from each other along the Y direction, where inner semiconductor fins **204** are interposed between two outer semiconductor fins **204'**. The outer semiconductor fins **204'** are identical to the inner semiconductor fins **204** in composition and structure but differ in their relative positions within the device region **203**. In the present embodiments, the outer semiconductor fins **204'** define two edges of the device region **203** (and thus may alternatively be referred to as edge semiconductor fins **204'**) along the X direction. In this regard, if a total number N of all the semiconductor fins disposed in the device region **203** is greater than or equal to two, then the number of outer semiconductor fins **204'** is two and the number of inner semiconductor fins **204** is (N-2). In one example, if N is four as depicted herein, then the number of the outer semiconductor fins **204'** is two and the number of the inner semiconductor fins **204** is also two. In another example, if N is two, then the number of the outer semiconductor fins **204'** is two and the number of the inner semiconductor fins **204** is zero, i.e., both of the semiconductor fins are considered outer semiconductor fins **204'**. In some embodiments, the device region **203** includes only one outer semiconductor fin **204'** (see, for example, FIGS. **17B**, **18A**, and **18B**). In other words, if N is less than or equal to two, then the semiconductor fin(s) in the device region **203** are all considered outer semiconductor fin(s) **204'**; otherwise, the device region **203** includes a mixture of outer semiconductor fins **204'** and inner semiconductor fins **204**.

(13) Referring to FIG. **2**, a top view of the device **200** depicts a portion of the substrate **202** over which the device region **203** is formed. Referring to FIG. **3**, which is a cross-sectional view of the device **200**, method **100** at operation **102** forms the semiconductor fins **204** and/or **204'** from the substrate **202**. The semiconductor fins **204** and/or **204'** may be fabricated using suitable processes including photolithography and etch processes. The photolithography process may include forming a photoresist layer (resist) overlying the substrate **202**, exposing the resist to a pattern, performing post-exposure bake processes, and developing the resist to form a masking element (not shown) including the resist. The masking element is then used for etching recesses into the substrate **202**, leaving the semiconductor fins **204** and/or **204'** on the substrate **202**. The etching process may include dry etching, wet etching, reactive ion etching (RIE), and/or other suitable processes.

(14) Numerous other embodiments of methods for forming the semiconductor fins **204** and/or **204'** may be suitable. For example, the semiconductor fins **204** and/or **204'** may be patterned using double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the semiconductor fins **204** and/or **204'**.

(15) The substrate **202** may include an elemental (single element) semiconductor, such as silicon (Si), germanium (Ge), and/or other suitable materials; a compound semiconductor, such as silicon

carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, indium antimonide, and/or other suitable materials; an alloy semiconductor, such as SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, and/or other suitable materials. The substrate **202** may be a single-layer material having a uniform composition. Alternatively, the substrate **202** may include multiple material layers having similar or different compositions suitable for IC device manufacturing.

(16) In some examples, various doped regions may be disposed in or on the substrate **202**. The doped regions may be doped with n-type dopants, such as phosphorus or arsenic, and/or p-type dopants, such as boron or BF₃, depending on design requirements. The doped regions may be formed directly on the substrate **202**, in a p-well structure, in an n-well structure, in a dual-well structure, or in a raised structure. Doped regions may be formed by implantation of dopant atoms, in-situ doped epitaxial growth, and/or other suitable techniques. Of course, these examples are for illustrative purposes only and are not intended to be limiting.

(17) In some embodiments, referring to FIGS. **4**, **5A**, and **5B**, the device region **203** is configured to provide one or more GAA FETs, where a multi-layer stack (ML) is formed over the substrate **202** and the semiconductor fins **204** and/or **204'** are subsequently formed from the ML and the substrate **202** at operation **102**. Referring to FIG. **4**, the ML includes alternating layers **204a** and **204b**, where the layer **204a** comprises a semiconductor material such as, for example, Si, Ge, SiC, SiGe, GeSn, SiGeSn, SiGeCSn, other suitable semiconductor materials, or combinations thereof, configured as channel regions of the GAA FET, and the layer **204b** is a sacrificial layer configured to be removed at a subsequent processing step and replaced by a gate structure. In some embodiments, each layer **204b** includes a semiconductor material different from the semiconductor material of the layer **204a**. In one such example, the layer **204a** may include elemental Si and the layer **204b** may include SiGe. In another example, the layer **204a** may include elemental Si, while the layer **204b** may include elemental Ge. The ML may be arranged with the layer **204b** directly contacting the substrate **202** and the layer **204a** subsequently disposed on the layer **204b** as depicted herein; alternatively, the ML may be arranged with the layer **204a** directly disposed on the substrate **202** and the layer **204b** disposed on the layer **204a**. In some examples, the ML may include a total of three to ten pairs of alternating layers **204a** and **204b**; of course, other configurations may also be applicable depending upon specific design requirements.

(18) In the present embodiments, forming the ML includes alternately growing the layers **204a** and **204b** in a series of epitaxy processes including chemical vapor deposition (CVD) techniques (for example, vapor-phase epitaxy (VPE), ultra-high vacuum CVD (UHV-CVD), low-pressure CVD (LP-CVD), and/or plasma-enhanced CVD (PE-CVD)), molecular beam epitaxy, other suitable selective epitaxial growth (SEG) processes, or combinations thereof. The epitaxy process may use gaseous and/or liquid precursors containing Si and/or Ge, which interact with the composition of the underlying substrate.

(19) Subsequently, referring to FIGS. **5A** and **5B**, method **100** forms one or more semiconductor fins **204** and/or **204'** from the ML and the substrate **202**. In the present embodiments, each semiconductor fin **204** or **204'** includes a base fin **204c** protruding from the substrate **202** and the stack of alternating layers **204a** and **204b** (i.e., the ML) disposed over the base fin **204c**. Methods of forming the semiconductor fins **204** and/or **204'** are discussed in detail above with respect to FIG. **3**. The semiconductor fins **204** and/or **204'** may be formed into various configurations depending upon desired design requirements. For example, the layers **204a** and **204b** may be formed into nanowires as depicted in FIG. **5A**, nanosheets as depicted in FIG. **5B**, or nanorods (not depicted). In some embodiments, referring to FIG. **5A**, a width **W1** of each nanowire measured along the Y direction is less than or equal to a separation distance **W2** between two adjacent semiconductor fins **204** and/or **204'**. In some embodiments, referring to FIG. **5B**, a width **W3** of each nanosheet measured along the Y direction is at least the same as the separation distance **W2** between adjacent semiconductor fins **204** and/or **204'**. A wire (or sheet) release process may then

remove the layers **204b** to form multiple openings between the layers **204a**, and metal gate structures are subsequently formed in the openings, thereby forming GAA FETs with the channel layers. For this reason, the layers **204a** are hereafter referred to as channel layers **204a**, and the layers **204b** are hereafter referred to as non-channel layers **204b**. It is noted that embodiments of method **100** disclosed herein are equally applicable to semiconductor fins **204** and/or **204'** having one channel layer (i.e., a uniform fin) as depicted in FIG. 3 and/or having multiple channel layers **204a** as depicted in FIGS. 5A and 5B. For illustrative purposes, subsequent operations **104** to **118** of method **100** are discussed in reference to the semiconductor fins **204** and/or **204'** having a uniform fin structure as depicted in FIG. 3.

(20) Now referring to FIG. 6, method **100** forms isolation structures **208** over the substrate **202**, thereby filling spaces between the semiconductor fins **204** and/or **204'**. The isolation structures **208** may include silicon oxide (SiO and/or SiO.sub.2), fluoride-doped silicate glass (FSG), a low-k dielectric material, and/or other suitable materials. The isolation structures **208** may include shallow trench isolation (STI) features. Other isolation structure such as field oxide, local oxidation of silicon (LOCOS), other suitable structures, or combinations thereof may also be implemented as the isolation structures **208**. The isolation structures **208** may be a single-layer structure or a multi-layer structure, for example, having one or more thermal oxide liner layers. The isolation structures **208** may be deposited by any suitable method, such as CVD, flowable CVD (FCVD), spin-on-glass (SOG), other suitable methods, or combinations thereof. Thereafter, one or more chemical mechanical planarization (CMP) processes are performed to the isolation structures **208** to planarize a top surface of the device **200** and to expose top surface of the semiconductor fins **204** and/or **204'**.

(21) Referring to FIGS. 7 and 8, method **100** at operation **106** forms a dielectric fin (or alternatively referred to as a vertical dielectric feature or a dielectric wall) **206** adjacent and substantially parallel to each outer semiconductor fin **204'**, i.e., bordering boundaries of the device region **203**, to protrude from the isolation structures **208**. Referring to FIG. 7, method **100** first patterns the isolation structures **208** to form trenches **209** therein. The trenches **209** may be formed by a series of patterning and etching processes. For example, a masking element (not depicted) including a photoresist layer may be first formed over the isolation structures **208** and patterned in a photolithography process to form a patterned masking element. Subsequently, portions of the isolation structures **208** exposed by the patterned masking element may be removed by one or more suitable etching processes to form the trenches **209**, after which the patterned masking element is removed from the device **200** by plasma ashing or resist stripping. In the depicted embodiments, the trenches **209** do not extend to expose the substrate **202**, i.e., the trenches **209** are embedded in the isolation structures **208**; however, the present disclosure is not limited to such configuration and may be applicable to embodiments in which the trenches **209** expose portions of the substrate **202**, such that the subsequently formed dielectric fins **206** vertically extend to contact the substrate **202**.

(22) Subsequently, referring to FIG. 8, a dielectric material is deposited in the trenches **209** by any suitable method, such as CVD, FCVD, ALD, other suitable methods, or combinations thereof, and planarized by one or more CMP processes to form the dielectric fins **206** in the isolation structures **208**. The dielectric fins **206** may include any suitable dielectric material such as, for example, silicon nitride (SiN), silicon oxide (SiO and/or SiO.sub.2), silicon carbide (SiC), carbon-containing silicon nitride (SiCN), carbon-containing silicon oxide (SiOC), oxygen-containing silicon nitride (SiON), carbon-and-oxygen-doped silicon nitride (SiOCN), a low-k dielectric material, hafnium oxide (HfO), aluminum oxide (AlO), other suitable dielectric materials, or combinations thereof. In the present embodiments, the composition of the dielectric material differs from that of the isolation structures **208** to ensure sufficient etching selectivity therebetween. Thereafter, referring to FIG. 9, the isolation structures **208** are recessed to expose portions of the semiconductor fins **204** and/or **204'** and the dielectric fins **206**. In the present embodiments, the isolation structures **208** are recessed by a suitable etching process without etching, or substantially etching, the semiconductor

fins **204** and/or **204'** and the dielectric fins **206**.

(23) In the present embodiments, two dielectric fins **206** are formed on opposite sides of each device region **203**. Stated differently, no dielectric fins **206** are present within the device region **203**. Stated in yet another way, the region disposed between two dielectric fins **206** is free of any additional dielectric fin **206**. For embodiments in which a single semiconductor fin (e.g., an outer semiconductor fin **204'** as depicted in FIGS. **17B**, **18A**, and **18B**) is provided in the device region **203**, one dielectric fin **206** is formed adjacent to each of the two sidewalls of the single semiconductor fin, i.e., the single semiconductor fin is disposed between two dielectric fins **206**. Furthermore, as depicted in FIG. **9**, each dielectric fin **206** is formed at a distance *D* away from the nearest semiconductor fin, where the distance *D* is determined based on the size of an epitaxial source/drain (S/D) feature to be formed in the semiconductor fin. In some embodiments, a dummy spacer (not depicted) may be formed between the dielectric fin **206** and an outer semiconductor fin **204'**, and the distance *D* is defined by the width of the dummy spacer. As will be discussed in detail below, the distance *D* is configured to allow the epitaxial S/D feature to contact the dielectric fins **206**, thereby enclosing an air gap below the epitaxial S/D feature.

(24) Now referring to FIGS. **10A-10C**, method **100** at operation **108** forms a dummy gate structure (i.e., a placeholder gate) **210** including polysilicon over channel regions of the semiconductor fins **204** and/or **204'**. FIGS. **10A** and **10B** illustrate a top view and a three-dimensional perspective view, respectively, of the device **200** after forming the dummy gate structure **210**. Cross-sectional views of the device **200** through S/D regions of the semiconductor fins **204** and/or **204'**, i.e., along the line AA', are depicted in FIGS. **11A**, **12A**, and **13-18B**, and cross-sectional views of the device **200** through channel regions of the semiconductor fins **204** and/or **204'**, i.e., along the line BB', are depicted in FIGS. **10C**, **11B**, **12B**, and **19A-20B**. In the present embodiments, portions of the dummy gate structure **210** are replaced with a metal gate structure after forming other components of the device **200**. The dummy gate structure **210** may be formed by a series of deposition and patterning processes. For example, the dummy gate structure **210** may be formed by depositing a polysilicon layer over the device region **203** and performing an etching process (e.g., a dry etching process) to remove portions of the polysilicon and form the dummy gate structure **210**. Though not depicted, the device **200** may include an interfacial layer formed over the semiconductor fins **204** and/or **204'** by a suitable method, such as thermal oxidation, chemical oxidation, other suitable methods, or combinations thereof, before depositing the polysilicon layer. The dummy gate structure **210** may further include other material layers including a dielectric layer, a hard mask layer, a diffusion layer, a capping layer, other suitable layers, or combinations thereof.

(25) Now referring to FIGS. **11A-12B**, method **100** at operation **110** forms gate spacers **211b** on sidewalls of the dummy gate structure **210**. Referring to FIGS. **11A** and **11B**, method **100** deposits a spacer layer **211** over the device **200**, such that the spacer layer **211** is formed conformally over the semiconductor fins **204** and/or **204'** and the dummy gate structure **210**. The spacer layer **211** may be a single-layer structure or a multi-layer structure and may include silicon nitride, silicon oxide (SiO and/or SiO.sub.2), silicon carbide (SiC), carbon-containing silicon nitride (SiCN), carbon-containing silicon oxide (SiOC), oxygen-containing silicon nitride (SiON), carbon-and-oxygen-doped silicon nitride (SiOCN), a low-k dielectric material, hafnium oxide (HfO), aluminum oxide (AlO), other suitable dielectric materials, or combinations thereof. In the present embodiments, the composition of the spacer layer **211** is different from the composition of the dielectric fins **206** to ensure etching selectivity between the gate spacers **211b** and the dielectric fins **206** when subjected to an etchant. In some embodiments, the dielectric fins **206** include a dielectric material having a dielectric constant that is lower than that of the spacer layer **211** in an effort to lower the parasitic capacitance of the device **200**. The spacer layer **211** may be formed by any suitable method, such as CVD, FCVD, ALD, PVD, other suitable methods, or combinations thereof. Subsequently, referring to FIG. **12B**, method **100** implements an anisotropic etching process (e.g., a dry etching process) to the spacer layer **211**, leaving portions of the spacer layer **211** on the sidewalls of the dummy gate

structure **210** as the gate spacers **211b**. In some embodiments, referring to FIG. **12A**, forming the gate spacers **211b** leaves portions of the spacer layer **211** on the sidewalls of the semiconductor fins **204** and/or **204'** and the dielectric fins **206** as fin spacers **211a**. In the present embodiments, an etchant is selected such that etching the spacer layer **211** does not etch, or does not substantially etch, the dielectric fins **206**. As depicted herein, the fin spacers **211a** are formed on bottom portions of the sidewalls of the semiconductor fins **204** and/or **204'** and the dielectric fins **206**.

(26) Referring to FIG. **13**, method **100** at operation **112** forms an S/D recess **213** in the S/D region of each of the semiconductor fins **204** and/or **204'**. In the present embodiments, method **100** forms the S/D recesses **213** by selectively etching the semiconductor fins **204** and/or **204'** without etching, or substantially etching, the dielectric fins **206** and the fin spacers **211a**. In some embodiments, method **100** implements a dry etching process that utilizes, for example, a chlorine-containing etchant including Cl.sub.2, SiCl.sub.4, BCl.sub.3, other chlorine-containing gas, or combinations thereof. Additionally or alternatively, other etching process(es) and etchant(s) may be utilized if the semiconductor fins **204** and/or **204'** include the ML as depicted in FIGS. **5A** and **5B**. In some embodiments, the etching process may be tuned by adjusting duration, temperature, pressure, source power, bias voltage, bias power, etchant flow rate, and/or other suitable parameters to control a depth of the S/D recesses **213**. A cleaning process may subsequently be performed to clean the S/D recesses **213** with a hydrofluoric acid (HF) solution or other suitable solution.

(27) For embodiments in which the semiconductor fins **204** and/or **204'** include the ML, i.e., configured to form a GAA FET, method **100** at operation **114** forms inner spacers (not depicted) on portions of the non-channel layers **204b** exposed in the S/D recesses **213**. The inner spacers may include any suitable dielectric material comprising silicon, carbon, oxygen, nitrogen, other elements, or combinations thereof. For example, the inner spacers may include silicon nitride (SiN), silicon oxide (SiO and/or SiO.sub.2), silicon carbide (SiC), carbon-containing silicon nitride (SiCN), carbon-containing silicon oxide (SiOC), oxygen-containing silicon nitride (SiON), silicon, carbon-and-oxygen-doped silicon nitride (SiOCN), a low-k dielectric material, tetraethylorthosilicate (TEOS), doped silicon oxide (e.g., borophosphosilicate glass (BPSG), fluoride-doped silicate glass (FSG), phosphosilicate glass (PSG), boron-doped silicate glass (BSG), etc.), air, other suitable dielectric material, or combination thereof. The inner spacers may each be configured as a single-layer structure or a multi-layer structure including a combination of the dielectric materials provided herein. In some embodiments, the inner spacers have a different composition from that of the gate spacers **211b** (and the fin spacers **211a**).

(28) Method **100** may form the inner spacers in a series of etching and deposition processes. For example, forming the inner spacers may begin with selectively removing portions of the non-channel layers **204b** without removing or substantially removing portions of the channel layers **204a** to form trenches. The non-channel layers **204b** may be removed by any suitable process, such as a dry etching process. Subsequently, one or more dielectric layers are formed in the trenches, followed by one or more etching processes to remove (i.e., etch back) excess dielectric layer(s) deposited on exposed surfaces of the channel layers **204a**, thereby forming the inner spacers on the non-channel layers **204b**. The one or more dielectric layers may be deposited by any suitable method, such as ALD, CVD, PVD, other suitable methods, or combinations thereof.

(29) Referring to FIG. **14**, method **100** at operation **116** forms the epitaxial S/D feature **214** in each S/D recess **213**. Each of the epitaxial S/D features **214** may be suitable for forming a p-type FET device (e.g., including a p-type epitaxial material) or alternatively, an n-type FET device (e.g., including an n-type epitaxial material). The p-type epitaxial material may include one or more epitaxial layers of silicon germanium (epi SiGe) doped with a p-type dopant such as boron, germanium, indium, other p-type dopants, or combinations thereof. The n-type epitaxial material may include one or more epitaxial layers of silicon (epi Si) or silicon carbon (epi SiC) doped with an n-type dopant such as arsenic, phosphorus, other n-type dopants, or combinations thereof. In some embodiments, the epitaxial S/D features **214** merge together as depicted herein; however, the

present embodiments are not limited as such.

(30) In some embodiments, one or more epitaxial growth processes are performed to grow an epitaxial material in each S/D recess **213**. For example, method **100** may implement an epitaxy growth process as discussed above with respect to forming the channel layers **204a** and the non-channel layers **204b** of the ML. In some embodiments, the epitaxial material is doped in-situ by adding a dopant to a source material during the epitaxial growth process. In some embodiments, the epitaxial material is doped by an ion implantation process after performing a deposition process. In some embodiments, an annealing process is subsequently performed to activate the dopants in the epitaxial S/D features **214**.

(31) In the present embodiments, the epitaxial S/D feature **214** formed adjacent to one of the dielectric fins **206** is configured to contact that dielectric fin **206**, thereby forming an outer air gap **230** near a bottom portion of the epitaxial S/D feature **214**. In other words, the epitaxial S/D features **214** formed at edges of the device region **203** laterally grow to touch a sidewall of their respective neighboring dielectric fin **206**. In the depicted embodiments, each outer air gap **230** is defined by the epitaxial S/D feature **214**, the dielectric fin **206**, and portions of the fin spacers **211a**. In some embodiments, two adjacent epitaxial S/D features **214** merge together to form an inner air gap **232** in a space between bottom portions of the epitaxial S/D features **214**. Each inner air gap **232** may be further defined by portions of the fin spacers **211a**. As discussed above with respect to FIG. **9**, the distance D between a dielectric fin **206** and its neighboring semiconductor fin (i.e., an outer semiconductor fin **204'**) is configured to allow the epitaxial S/D feature **214** to grow laterally until it contacts the sidewall of the dielectric fin **206**. Therefore, in the present embodiments, a lateral dimension (i.e., the dimension measured along the Y direction)—and therefore the volume—of the outer air gap **230** depends on the magnitude of the distance D.

(32) A general strategy for improving performance (e.g., processing speed) of an FET is to reduce the device's parasitic capacitance, which lowers the RC delay of the device. Reducing parasitic capacitance may be implemented by incorporating materials with lower dielectric constant, such as air in the form of an air gap, into the device's structure. While existing methods of introducing air gap(s) in an FET have generally adequate, they have not been entirely satisfactory in all aspects. For example, when a device includes two or more semiconductor fins (active regions), parasitic capacitance of the FET may be reduced by merging neighboring S/D features formed over the semiconductor fins to create inner air gap(s) below the S/D features, such as the inner air gaps **232** provided herein. However, when devices continue to decrease in size, the number of semiconductor fins within a given device may be reduced to less than two, thus preventing the formation of any inner air gap. The present embodiments provide methods of forming additional air gaps, such as the outer air gaps **230** provided herein, adjacent to the outermost semiconductor fins, such that at least two air gaps (i.e., the outer air gaps **230**) are configured for each device regardless of how many semiconductor fins are present. In the present embodiments, incorporating dielectric fins adjacent to the outer semiconductor fins increases the total number of air gaps in a device by two, which is especially advantageous for embodiments in which only one semiconductor fin is present. In some embodiments, the total number of air gaps formed within a device exceeds the number of semiconductor fins present, thereby maximizing the capacitance-reducing effect of the air gaps.

(33) Referring to FIGS. **15A** and **15B**, method **100** at operation **118** forms an etch-stop layer (ESL) **220** over the device **200**. The ESL **220** may include silicon nitride (SiN), carbon-containing silicon nitride (SiCN), oxygen-containing silicon nitride (SiON), silicon, carbon-and-oxygen-doped silicon nitride (SiOCN), other suitable materials, or combinations thereof. In some embodiments, the dielectric fins **206** include a dielectric material having a dielectric constant that is lower than that of the fin spacers **211a** and/or the ESL **220** in an effort to lower the parasitic capacitance of the device **200**. In the present embodiments, the ESL **220** is formed conformally over the device **200** by CVD, PVD, ALD, other suitable methods, or combinations thereof. Referring to FIG. **15A**, the ESL **220** is formed over top surface of the merged epitaxial S/D features **214** and the dielectric fins **206** but

does not fill the air gap **230** or the air gap **232**. Alternatively, referring to FIG. **15B**, in addition to being formed over the top surface of the merged epitaxial S/D features **214** and the dielectric fins **206**, the ESL **220** may partially fill the air gap **230** and the air gap **232**, thereby reducing, though not eliminating, volume of each air gap. Such partial filling of the air gap **230** and the air gap **232** by the ESL **220** may occur when the epitaxial S/D features **214** only merge slightly, which leads to enlarged volume of the air gap **230** and the air gap **232**.

(34) Subsequently, referring to FIGS. **16A** and **16B** which correspond to FIGS. **15A** and **15B**, respectively, method **100** at operation **118** forms an interlayer dielectric (ILD) layer **222** over the ESL **220**. The ILD layer **222** may include silicon oxide (SiO and/or SiO.sub.2), a low-k dielectric material, TEOS, doped silicon oxide (e.g., BPSG, FSG, PSG, BSG, etc.), other suitable dielectric materials, or combinations thereof, and may be formed by CVD, FCVD, SOG, other suitable methods, or combinations thereof. Thereafter, method **100** may planarize the ILD layer **222** in one or more CMP processes to expose a top surface of the dummy gate structure **210**. FIGS. **16C** and **16D** depict embodiments of the device **200** similar to those of FIGS. **16A** and **16B** with the exception that the semiconductor fins **204** and/or **204'** each include the channel layers **204a** interleaved with non-channel layers (not depicted) as depicted in FIGS. **5A** and **5B** and configured to form GAA FET(s).

(35) In the present embodiments, a number M of the outer air gap **230** is consistent with the number of dielectric fins **206**, which is two regardless of the number N of the semiconductor fins **204** and/or **204'**. A number P of the inner air gaps **232**, on the other hand, depends upon a number N' of the epitaxial S/D features **214** merged together, e.g., if N' is zero, then no inner air gaps **232** are formed, $P=0$; otherwise, $P=N'-1$. Accordingly, in the absence of any dielectric fins **206**, merging two adjacent epitaxial S/D features **204** would yield no outer air gaps **230**, or $M=0$, and no more than a total of $(N'-1)$ inner air gaps **232**, or $(M+P) \leq (N'-1)$. Embodiments provided herein are configured to reduce the parasitic capacitance of the device **200** by maximizing the total number $(M+P)$ of the air gaps with respect to the number N of the semiconductor fins **204** and/or **204'** in the device region **203**. In one example, referring to FIGS. **14-16B**, because the device **200** includes two dielectric fins **206** and four merged epitaxial S/D features **214** and/or **204'**, or $N=N'=4$, there are two outer air gaps **230** ($M=2$) and three inner air gaps **232**, or $P=N'-1=3$, and the total number of air gaps is $M+P=5$, which is greater than N . In a similar example, referring to FIG. **17A**, because the device **200** includes two dielectric fins **206** and two merged epitaxial S/D features **214**, $N=N'=2$, there are two outer air gaps **230** ($M=2$) and one inner air gap **232** ($P=1$), and the total number of air gaps is $M+P=3$, which greater than N .

(36) For embodiments in which the device **200** includes a single semiconductor fin **204'** ($N=1$), referring to FIGS. **17B**, **18A**, and **18B**, no inner air gap **232** is formed, or $P=0$, and the total number of air gaps therefore arises solely from the outer air gaps **230** formed by the single semiconductor fin **204'** and the two dielectric fins **206**. In this regard, the total number of air gaps is $M+P=2$, which is greater than N . In the absence of any dielectric fin **206**, however, no air gap would be formed to reduce the parasitic capacitance of the device **200**, or $M+P=0$. Thus, the dielectric fins **206** reduces the parasitic capacitance of a device by providing two additional outer air gaps with the outer semiconductor fins **204'**, and the effect of such reduction is especially beneficial when the device includes a single semiconductor fin **204'**.

(37) It is noted that the formation of the outer air gaps **230** is not dependent on specific configurations of the semiconductor fins **204** and/or **204'**. In other words, the semiconductor fins **204** and/or **204'** may be configured to form FinFETs (as depicted in FIGS. **17A** and **17B**) and/or GAA FETs (as depicted in FIGS. **18A** and **18B**). Furthermore, embodiments provided herein may be applicable to various configurations of the GAA FETs, such as nanosheet-based GAA FET as depicted in FIG. **18A** and nanorod-based GAA FET as depicted in FIG. **18B**. Still further, the present embodiments do not limit the width $W_{sub.S}$ of the semiconductor fins **204** and **204'** or the width $W_{sub.D}$ of the dielectric fins **206** to specific values. For example, referring to FIGS. **17A**,

17B, and 18B, W.sub.S is less than W.sub.D, and referring to FIG. 18A, W.sub.S is greater than W.sub.D. Of course, W.sub.S may be substantially the same as W.sub.D according to some embodiments (not depicted).

(38) Now referring to FIG. 19A, method 100 at operation 120 removes the dummy gate structure 210 to form a gate trench 226 by any suitable etching process, such as a dry etching process, thereby exposing channel regions of the semiconductor fins 204 and/or 204'. In some embodiments, the interfacial layer, if present, remains over the semiconductor fins 204 and/or 204' in the gate trench 226.

(39) For embodiments in which the semiconductor fins 204 and/or 204' include the ML configured to form GAA FETs, referring to FIG. 19B, method 100 proceeds from operation 120 to operation 122 to remove the non-channel layers 204b from the ML, thereby forming openings 228 interleaved with the channel layers 204a. Method 100 selectively removes the non-channel layers 204b without removing, or substantially removing, the channel layers 204a by any suitable etching process, such as dry etching, wet etching, RIE, or combinations thereof. In one example, a wet etching process employing ammonia (NH.sub.3) and/or hydrogen peroxide (H.sub.2O.sub.2) may be performed to selectively remove the non-channel layers 204b. In another example, a dry etching process employing HF and/or other fluorine-based etchant(s), such as CF.sub.4, SF.sub.6, CH.sub.2F.sub.2, CHF.sub.3, C.sub.2F.sub.6, other fluorine-containing etchants, or combinations thereof, may be implemented to remove the non-channel layers 204b.

(40) Subsequently, referring to FIGS. 20A and 20B, which correspond to FIGS. 19A and 19B, respectively, method 100 at operation 124 forms a metal gate structure 240 over the channel region of each of the semiconductor fins 204 and/or 204', i.e., in the gate trench 226 and, if applicable, in the openings 228, where the metal gate structure 240 includes at least a gate dielectric layer 242 and a metal gate electrode 244 disposed over the gate dielectric layer 242. For embodiments in which the semiconductor fins 204 and/or 204' include the ML configured to form GAA FETs, referring to FIG. 20B, the gate dielectric layer 242 wraps around each channel layer 204a, and the metal gate electrode 244 is configured to fill in the openings 228 between the channel layers 204a as well as in the gate trench 226.

(41) In the present embodiments, the gate dielectric layer 242 includes any suitable high-k (i.e., having a dielectric constant greater than that of silicon oxide, which is about 3.9) dielectric material, such as hafnium oxide, lanthanum oxide, other suitable materials, or combinations thereof. The metal gate electrode 244 may include at least one work function metal layer (not depicted separately) and a bulk conductive layer (not depicted separately) disposed thereover. The work function metal layer may be a p-type or an n-type work function metal layer. Example work function materials include TiN, TaN, Ru, Mo, Al, WN, ZrSi.sub.2, MoSi.sub.2, TaSi.sub.2, NiSi.sub.2, Ti, Ag, TaAl, TaAlC, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, other suitable work function materials, or combinations thereof. The bulk conductive layer may include Cu, W, Al, Co, Ru, other suitable materials, or combinations thereof. The metal gate structure 240 may further include numerous other layers (not depicted), such as an interfacial layer, a capping layer, a barrier layer, other suitable layers, or combinations thereof. Various layers of the metal gate structure 240 may be deposited by any suitable method, such as chemical oxidation, thermal oxidation, ALD, CVD, PVD, plating, other suitable methods, or combinations thereof.

(42) Thereafter, method 100 at operation 126 may perform additional processing steps to the device 200. For example, method 100 may form S/D contacts (not depicted) over the S/D features 214, where each S/D contact may include any suitable conductive material, such as Co, W, Ru, Cu, Al, Ti, Ni, Au, Pt, Pd, other suitable conductive materials, or combinations thereof. Method 100 may form an S/D contact opening in the ILD layer 222 via a series of patterning and etching processes and subsequently deposit a conductive material in the S/D contact opening using any suitable method, such as CVD, ALD, PVD, plating, other suitable processes, or combinations thereof. In some embodiments, a silicide layer (not depicted) is formed between the S/D features 214 and the

S/D contact. The silicide layer may include nickel silicide, cobalt silicide, tungsten silicide, tantalum silicide, titanium silicide, platinum silicide, erbium silicide, palladium silicide, other suitable silicide, or combinations thereof. The silicide layer may be formed over the device **200** by a deposition process such as CVD, ALD, PVD, or combinations thereof. Subsequently, method **100** may form additional features over the device **200** such as, for example, a gate contact (not depicted) configured to be coupled with the metal gate structure **240**, vertical interconnect features (e.g., vias; not depicted), horizontal interconnect features (e.g., conductive lines; not depicted), additional intermetal dielectric layers (e.g., ESLs and ILD layers; not depicted), other suitable features, or combinations thereof.

(43) Although not intended to be limiting, one or more embodiments of the present disclosure provide many benefits to a semiconductor device and the formation thereof. For example, in the present embodiments, parasitic capacitance of a device (e.g., a FinFET, a GAA FET, etc.) is reduced by forming vertical dielectric features (e.g., dielectric fins) adjacent to epitaxial S/D features of the device. In some embodiments, forming the dielectric fins introduces air gaps at bottom of those semiconductor fins disposed at outer edges of a device region that includes an array of semiconductor fins. As provided herein, introducing air gap to the outer (or edge) semiconductor fins is important for reducing parasitic capacitance of the device when efforts of decreasing the number of semiconductor fins (also known as fin depopulation) are implemented. In some embodiments of the present disclosure, the number of semiconductor fins may be reduced to two or less. Embodiments of the disclosed methods can be readily integrated into existing processes and technologies for manufacturing various FETs.

(44) In one aspect, the present embodiments provide a semiconductor structure that includes a semiconductor fin protruding from a substrate, an S/D feature disposed over the semiconductor fin, and a first dielectric fin and a second dielectric fin disposed over the substrate, where the semiconductor fin is disposed between the first dielectric fin and the second dielectric fin, where a first air gap is enclosed by a first sidewall of the epitaxial S/D feature and the first dielectric fin, and where a second air gap is enclosed by a second sidewall of the epitaxial S/D feature and the second dielectric fin.

(45) In another aspect, the present embodiments provide a semiconductor structure that includes a first semiconductor fin and a second semiconductor fin disposed over a substrate and oriented lengthwise along a first direction, a gate stack disposed over a first channel region of the first semiconductor fin and a second channel region of the second semiconductor fin and oriented lengthwise along a second direction substantially perpendicular to the first direction, a first S/D feature and a second S/D feature disposed over the first semiconductor fin and the second semiconductor fin, respectively, and a first dielectric fin and a second dielectric fin disposed over the substrate and oriented lengthwise along the first direction, where the first semiconductor fin and the second semiconductor fin are disposed between the first dielectric fin and the second dielectric fin, where the first dielectric fin forms a first air gap with the first S/D feature, and where the second dielectric fin forms a second air gap with the second S/D feature. In the present embodiments, the first channel region and the second channel region each includes a plurality of semiconductor layers interleaved with the gate stack.

(46) In yet another aspect, the present embodiments include forming a semiconductor fin protruding from a substrate, forming a dielectric fin adjacent to the semiconductor fin, where the dielectric fin is oriented substantially parallel to the semiconductor fin, removing a portion of the semiconductor fin to form an S/D recess, and forming an S/D feature in the S/D recess, such that the S/D feature contacts a sidewall of the dielectric fin, thereby defining an air gap.

(47) The foregoing outlines features of several embodiments so that those of ordinary skill in the art may better understand the aspects of the present disclosure. Those of ordinary skill in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the

same advantages of the embodiments introduced herein. Those of ordinary skill in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a plurality of nanostructures over a substrate; a gate structure wrapping around and over at least a portion of one of the plurality of nanostructures; a dielectric fin over the substrate, wherein a top surface of the dielectric fin is coplanar with a top surface of a topmost nanostructure of the plurality of nanostructures; and a source/drain feature adjacent to the gate structure and coupled to the at least one of the plurality of nanostructures, wherein an air gap is disposed between the source/drain feature and the dielectric fin.
2. The semiconductor structure of claim 1, further comprising: gate spacers extending along sidewalls of the gate structure; and fin sidewall spacers extending along sidewalls of the dielectric fin and a bottom portion of the source/drain feature, wherein the gate spacers and the fin sidewall spacers have a same composition.
3. The semiconductor structure of claim 1, wherein the source/drain feature is a first source/drain feature, the air gap is a first air gap, the semiconductor structure further comprising: a second source/drain feature next to the first source/drain feature; and a second air gap disposed between the first source/drain feature and the second source/drain feature.
4. The semiconductor structure of claim 3, wherein the first air gap spans a first width, the second air gap spans a second width greater than the first width.
5. The semiconductor structure of claim 1, further comprising: a first dielectric layer conformally extending along over top surfaces of the dielectric fin and the source/drain feature; and a second dielectric layer on the first dielectric layer.
6. The semiconductor structure of claim 5, wherein the air gap is partially defined by the first dielectric layer.
7. The semiconductor structure of claim 1, wherein the dielectric fin is a first dielectric fin, the semiconductor structure further comprises: a second dielectric fin; and another air gap disposed between the source/drain feature and the second dielectric fin.
8. The semiconductor structure of claim 1, further comprising: an isolation feature over the substrate, wherein the dielectric fin extends into the isolation feature.
9. The semiconductor structure of claim 1, wherein the dielectric fin extends lengthwise along a first direction, and the gate structure extends lengthwise along a second direction substantially perpendicular to the first direction.
10. The semiconductor structure of claim 1, wherein the plurality of nanostructures are disposed directly over a portion of the substrate, the portion of the substrate has a width different than a width of the dielectric fin.
11. A semiconductor structure, comprising: a first source/drain feature coupled to a first channel region; a second source/drain feature coupled to a second channel region; a gate structure over the first channel region and the second channel region; a first sidewall spacer extending along a bottom portion of the first source/drain feature; a second sidewall spacer extending along a bottom portion of the second source/drain feature; and an air gap extending from the first sidewall spacer to the second sidewall spacer.
12. The semiconductor structure of claim 11, further comprising: gate spacers extending along sidewalls of the gate structure; wherein the gate spacers, the first sidewall spacer, and the second sidewall spacer have a same composition.
13. The semiconductor structure of claim 12, further comprising: a first dielectric layer conformally extending along over top surfaces of the first and second source/drain features; and a second

dielectric layer on the first dielectric layer.

14. The semiconductor structure of claim 13, wherein a top surface of the second dielectric layer is coplanar with top surfaces of the gate spacers.

15. The semiconductor structure of claim 11, further comprising: a dielectric fin disposed adjacent to the first source/drain feature, and a distance between the dielectric fin and the first source/drain feature is less than a distance between the first source/drain feature and the second source/drain feature.

16. The semiconductor structure of claim 15, further comprising: another air gap disposed between the first source/drain feature and the dielectric fin.

17. A semiconductor structure, comprising: an isolation feature over a substrate; a dielectric fin over the substrate and extending into the isolation feature, wherein the dielectric fin comprises a first sidewall and a second sidewall being opposite the first sidewall, wherein the first sidewall comprises: a bottom portion next to the isolation feature, a lower portion over the bottom portion and adjacent to a spacer layer, an upper portion over the lower portion and being exposed to an air gap; and a top portion over the upper portion and adjacent to a dielectric layer having a composition different than the dielectric fin and the isolation feature.

18. The semiconductor structure of claim 17, further comprising: a source/drain feature adjacent to the dielectric fin, wherein the air gap is disposed between the source/drain feature and the dielectric fin.

19. The semiconductor structure of claim 18, wherein the dielectric layer further extends along a top surface of the dielectric fin and a sidewall surface of the source/drain feature.

20. The semiconductor structure of claim 18, further comprising: a gate structure intersecting with the dielectric fin and adjacent to the source/drain feature; and a gate spacer extending along a sidewall surface of the gate structure, wherein the gate spacer and the spacer layer have a same composition.
